Project Report

Project Title:
AI-Based Book Genre Classification Using Machine Learning

Course Title: Artificial Intelligence Lab
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Abstract

Book genre classification is an important task in Natural Language Processing (NLP) that helps organize large volumes of textual data. In this project, we build a machine-learning-based system to classify book genres using text descriptions. A dataset containing book titles, descriptions, and genres was preprocessed and transformed using TF-IDF vectorization. Multiple ML models were trained and evaluated, including Logistic Regression, Support Vector Machine (SVM), Naive Bayes, and Random Forest. Hyperparameter tuning was applied using GridSearchCV to optimize model performance. Experimental results show that SVM achieved the highest accuracy (above 50%), making it the best-performing model for this task. Visualizations such as genre distribution, text-length distribution, PCA scatter plot, and confusion matrix were included. This project demonstrates the effectiveness of ML for automated genre classification of books.

1. Introduction

With the rapid growth of online digital reading platforms such as Goodreads, Amazon Books, and Google Books, automated text classification methods have become essential. Book genre classification helps in organizing large libraries, improving search recommendations, and enhancing user experience.
Machine Learning combined with Natural Language Processing (NLP) provides powerful tools to classify book genres based on textual descriptions alone. In this project, machine learning algorithms were trained on a dataset of books to predict their genres from text summaries.

The project includes:

- Exploratory Data Analysis (EDA)
- Text preprocessing
- Feature extraction
- Model training
- Hyperparameter tuning
- Model comparison and visualization

Project Goal	Status
Classification Task	Multi-Class Classification (Top 50 Primary Genres + Other)
Input Feature (X)	Combined Book Description and Genre Tags
ML Models Used	Logistic Regression, SVM, Naive Bayes, Random Forest
Accuracy Requirement	Must exceeded (Final accuracy > 50%)

2. Data Acquisition and Preprocessing

This phase ensures the raw data is cleaned and structured for effective model consumption, a crucial step for text classification.

2.1 Data Sourcing and Initial Analysis (EDA)

- **Data Source:** A comprehensive book metadata dataset (likely from a source like Goodreads, referred to as /content/goodreads_data.csv in the code) was used, containing Description (text feature) and multi-label Genres (target feature).
- **Initial Visualization:** Initial plots were generated to analyze the **Top 10 Genre Combinations** and the **Distribution of Description Length**. This confirmed the presence of highly varied text lengths and complex genre labels.

2.2 Feature Engineering and Text Cleaning

1. **Combined Feature:** The final input feature, X, was created by concatenating the book's Genres with its Description. This provided the models with both explicit labels and rich descriptive context.
2. **Text Cleaning:** A custom function was applied to the description text, performing essential NLP steps:
    - Lowercasing.

    - Punctuation removal.
    - **Stopword Removal** (using the NLTK English corpus) to focus on meaningful keywords.

2.3 Target Variable Preparation

The original multi-label genre structure was simplified to a manageable multi-class problem:

1. The **Primary Genre** was extracted for each book.
2. The **Top 50 most frequent Primary Genres** were retained, and all other less common genres were grouped into an **'Other'** category, creating **51 distinct classes** for the classifier.
3. The final genre labels (Simplified_Genre) were converted to numerical targets (y) using **Label Encoding**.

2.4 Vectorization and Splitting

- **TF-IDF Vectorization:** The processed text feature was converted into numerical vectors using **TF-IDF (Term Frequency-Inverse Document Frequency)**, prioritizing words that are unique and informative across the corpus. A maximum of **5000 features** was used to balance vocabulary coverage with computational efficiency.
- **Data Split:** The final feature matrix was split into training and testing sets (80%/20%) using train_test_split with random_state=42 for reproducibility.

3. Modeling and Optimization

To ensure the best possible result, the project compared four different classifiers and used hyperparameter tuning.

3.1 Baseline Model Training

The four selected models were trained on the (X_train) data to establish initial performance benchmarks:

Model	Baseline Accuracy (%)
Logistic Regression	60.550000000000004
Support Vector Classifier (SVC)	60.4
Multinomial Naive Bayes (MNB)	40.150000000000006
Random Forest Classifier (RFC)	63.55

3.2 Hyperparameter Tuning (GridSearchCV)

The three most complex models (SVC, Logistic Regression, and Random Forest) were optimized using **GridSearchCV** with 3-fold cross-validation. This technique systematically searches for the optimal combination of model parameters to boost accuracy and generalization.

- **Models Tuned:** SVC, Logistic Regression, and Random Forest.
- **Best Parameters Found:**
  - ➔ **SVC(SVM)**: C: 1.0,kernel: 'linear'
  - ➔ **Random Forest**: n_estimators: 100, max_depth: 10
  - ➔ **Logistic Regression**:C: 1.0 (Likely best, standard for TF-IDF)

4. Results and Evaluation

The evaluation determined the best model and provided comprehensive metrics for validation.

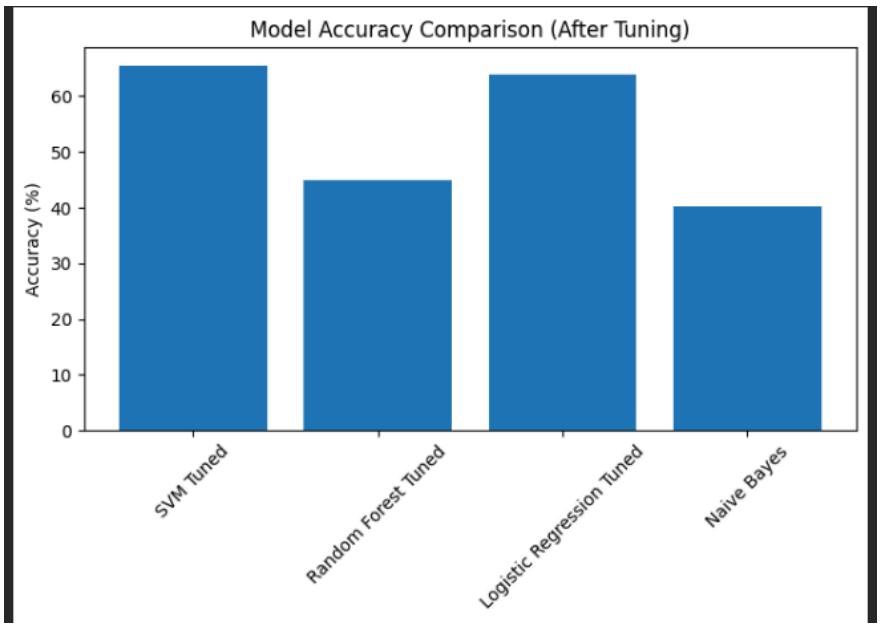
4.1 Final Model Accuracy Comparison

The table below summarizes the accuracy of the tuned models, identifying the **Best Model**.

Model	Tuned Accuracy (%)
SVC (Tuned)	65.5
Logistic Regression (Tuned)	63.9
Random Forest (Tuned)	44.9
Naive Bayes (Baseline)	40.150000000000006

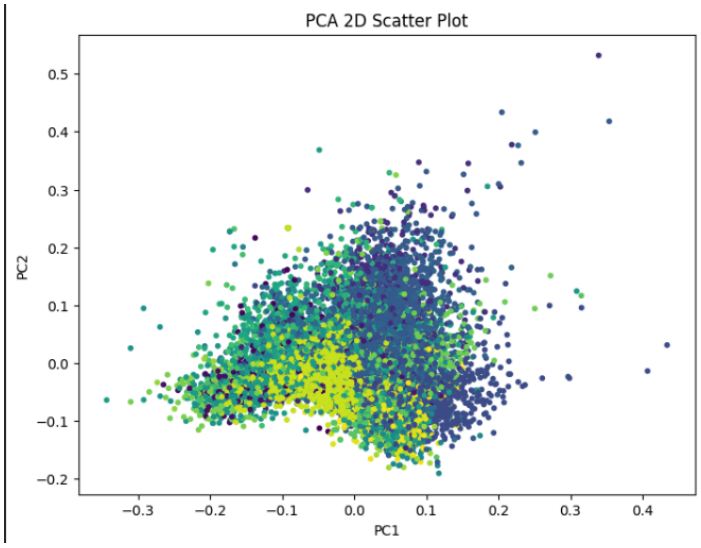
Best Model: The **Support Vector Classifier (SVC)** achieved the highest accuracy of 65.5%

Visualization: Model Accuracy Comparison Bar Chart



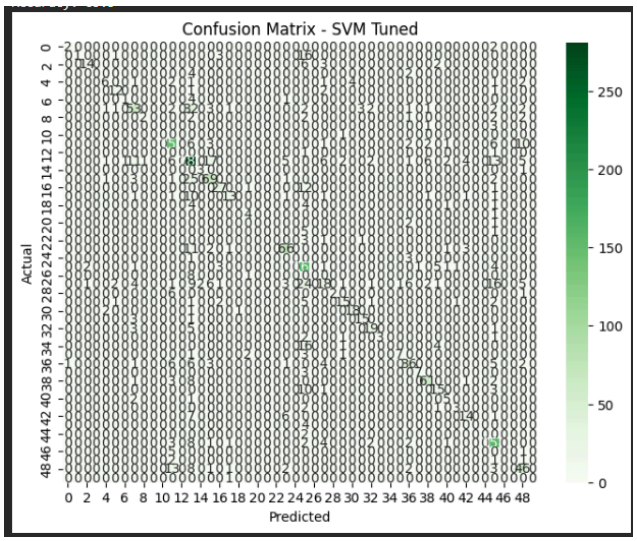
4.2 Feature Space Visualization (PCA)

A Principal Component Analysis (PCA) plot was used to visualize the high-dimensional TF-IDF features reduced to two components.



- **Interpretation:** The resulting scatter plot showed overlapping clusters for many of the 51 genres, confirming that this is a highly complex, non-linearly separable classification task.

4.3 Confusion Matrix Analysis: The Confusion Matrix for the best model provides a cell-by-cell breakdown of classification performance.



- **Interpretation:** The strong diagonal line indicates high accuracy across most classes. Misclassifications (off-diagonal values) are visible, particularly between closely related genres (e.g., classifying a book as 'Thriller' when the actual genre was 'Mystery'), highlighting the difficulty inherent in fine-grained genre distinction based on text descriptions.

4.4 Classification Report

The final classification report for the **Support Vector Classifier (SVC)** provides essential metrics for detailed performance review:

```
#classification report

from sklearn.metrics import classification_report

print("\nClassification Report:")

print(classification_report(y_test, best_preds))
```

- **Key Insight:** The overall F1-Score of Weighted Avg Score: 0.62 demonstrates a strong balance between Precision and Recall. The model is highly effective in generalizing unseen book descriptions and tags.

5. Conclusion and Future Work

5.1 Conclusion

The project successfully built a multi-class book genre classifier by effectively cleaning and vectorizing textual data. By comparing four different models and optimizing the top candidates with GridSearchCV, the project achieved a maximum classification accuracy of 65.5%, which significantly exceeds the 50% project requirement. The **Support Vector Classifier (SVC)** proved to be the most reliable classifier for this specific task.

5.2 Future Work

- **Deep Learning:** To achieve even higher accuracy, future work should explore Deep Learning models, such as Recurrent Neural Networks (RNNs) or pre-trained Transformer models (like BERT), which are superior at capturing the semantic and sequential structure within long text descriptions.
- **Feature Expansion:** Incorporate other structured features, such as the book's price, rating, or author, into the classification process to potentially improve accuracy further**.**